

RAJIV RANJAN JHA

Full-Stack AI Engineer — GenAI, Agentic Systems & Production ML

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SUMMARY

AI Engineer with 4 years progressing from forecasting models to full-stack GenAI platforms. Architected multi-agent RAG systems delivering **<800ms TTFT at 95%+ groundedness**, built parameter-efficient fine-tuning pipelines on consumer GPUs, and owned end-to-end delivery from Medallion ETLs through Cloud Run deployments.

Proven cross-layer ownership: **data engineering → model training → API serving → infrastructure.**

TECHNICAL SKILLS

GenAI / LLMs: Google ADK, MCP Tools, RAG (GraphRAG, Schema-RAG), Agentic Workflows, LoRA/QLoRA (PEFT), Unsloth, Hugging Face, Ollama, Sentence-Transformers, Prompt Engineering

ML / Data Science: LightGBM, Time-Series Forecasting (WAPE), YOLO, ArcFace, Diffusion Models, CPT/SFT Fine-Tuning, Statistical Modeling, Evaluation Frameworks

Data Engineering: BigQuery, PostgreSQL / AlloyDB, Medallion Architecture (Bronze/Silver/Gold), Dataform, ETL Pipelines, Data Curation (Dedup, PII Redaction, Quality Scoring)

Backend / Serving: FastAPI, Django (Async I/O), Redis Caching, REST APIs, SSE Streaming, Pydantic, Asyncio, React, Next.js

MLOps / Infrastructure: Docker, Google Cloud Run, Azure / Databricks, AI Studio, CI/CD, Model Registry, Pipeline Orchestration, Cost Optimization

EXPERIENCE

Senior Data Scientist / AI Engineer — Alkye Technologies

Sep 2024 - Present

Merchmix — AI-Powered SaaS Inventory Planner

- Eliminated 70–80% of analyst queries by architecting a multi-modal RAG chatbot with Google ADK and MCP tools over BigQuery/PostgreSQL, achieving **<800ms TTFT, >95% groundedness, and 85%+ retrieval recall@5**.
- Cut dashboard query costs **60%** by building a Medallion (Bronze/Silver/Gold) pipeline replicating BigQuery into AlloyDB via API/POST ingestion with zero data-freshness loss.
- Delivered **86% WAPE** forecast accuracy by training LightGBM time-series models over brand/category hierarchies for weekly and daily SKU-level demand planning.
- Reduced API response latency under load by migrating Django endpoints from sync to async I/O with Redis page caching, load-balanced serving, and CI/CD-driven Docker/Cloud Run deployments.

Fittora — AI-Powered Made-to-Order Tailoring Platform (Australian Client)

- Achieved **95%+ identity-retrieval accuracy** (ArcFace cosine similarity, matching published SOTA 94–99%) by solving multi-angle identity drift via custom annotation and model fine-tuning.
- Cut mockup latency to **<1s (3–5× faster than baseline diffusion)** by designing a dual-path architecture (primary face-swap + fallback model), holding **99%+ generation success rate**.

Data Scientist — SK Associates

Jul 2022 - Sep 2024

- Deployed custom YOLO object-detection models end-to-end (annotation → training → serving) for client computer-vision applications.
- Replaced raw-table reporting with executive dashboards surfacing revenue and profit drivers for banking-client leadership decisions.

FLAGSHIP PROJECTS

LLMOps Fine-Tuning Pipeline — E-Commerce Domain Adaptation | Python · PyTorch · Unsloth · LoRA/PEFT · FastAPI · SQLite

- Domain-adapted TinyLlama (1.1B params) on a single 8GB consumer GPU using LoRA (rank-32, rsLoRA) with **2× throughput** via Unsloth BF16 acceleration and gradient checkpointing.
- Built a 7-filter data-curation engine (PII redaction, MinHash dedup, quality scoring, percentile pruning), reducing 14K raw records to 6.4K clean documents (**54% rejection**) with full audit trails.
- Shipped a production-grade FastAPI service with SQLite job management, SSE log streaming, and a dark/light Web UI — domain-portable via YAML config with zero code changes.

Autonomous Coding Agent — Evidence-Gated Execution Harness | Python · Asyncio · Vertex AI (Gemini) · Playwright · Pydantic · 324 Tests

- Built an autonomous agent with 15 typed tools that plans, edits files, executes shell commands, and verifies work via a real browser's accessibility tree, with no LLM awareness of the underlying tool layer.
- Prevented false-success reporting with evidence gates that independently re-execute success predicates, catching premature completions in **3 of 9 (33%)** successful runs.
- Compressed tool output **~94%** via an observation compiler while preserving error spans, holding working context at **~22K tokens** against a 1M window — total inference spend under **\$0.50**.

EDUCATION

B.Tech — CV Raman Global University, Bhubaneswar

Jul 2018 - Jun 2022